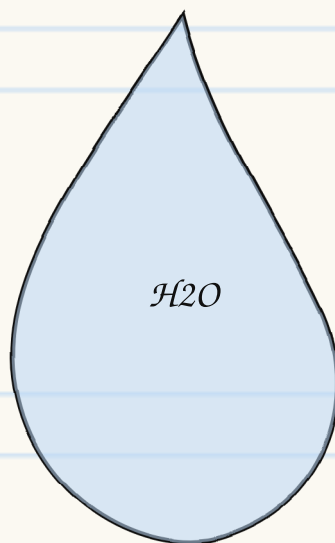


Engineering Chemistry

BT-101 | Unit 1

Water Chemistry

Hardness | EDTA Method | Alkalinity | Numericals



Topics Covered:

- *Definition of Hardness, Temporary & Permanent Hardness*
- *EDTA Method for Determination of Hardness*
- *Alkalinity - Meaning and Determination*
- *Units of Hardness*
- *Numerical Problems on Hardness & Alkalinity*

Handwritten style revision notes - for quick exam preparation

1. Hardness of Water

Definition: Hardness is that property of water which does **not allow soap to lather (form foam) easily**. It is caused due to presence of **dissolved salts of Calcium (Ca^{2+}) and Magnesium (Mg^{2+})** in water, mainly as bicarbonates, chlorides and sulphates.

* Water which gives lather easily with soap is called **Soft Water**.

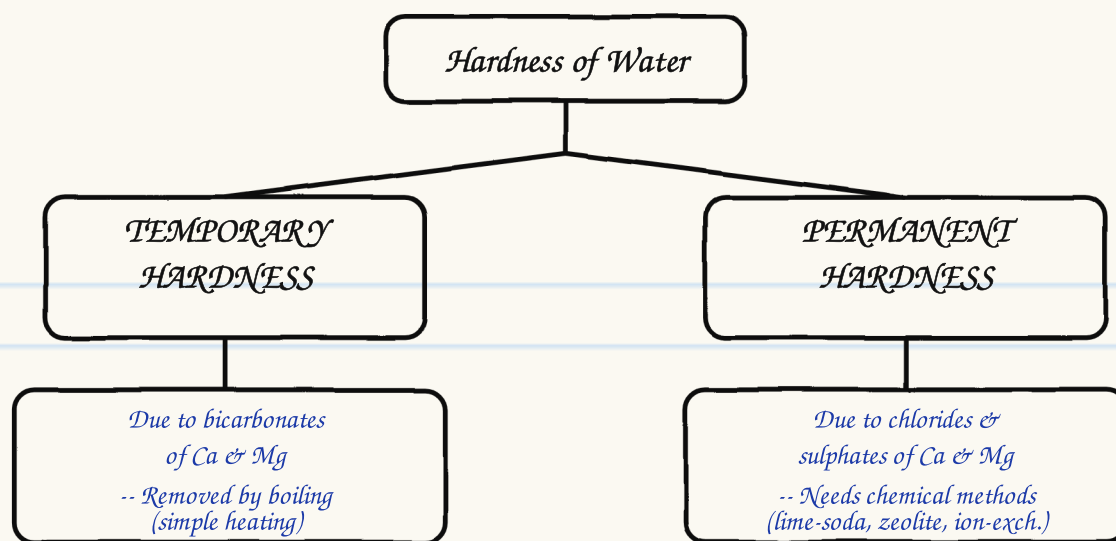
Water which does not give lather easily is called **Hard Water**.

Why does hard water not lather?

Soap is sodium/potassium salt of long chain fatty acid (say sodium stearate). When soap reacts with $\text{Ca}^{2+}/\text{Mg}^{2+}$ ions present in hard water, an insoluble sticky **precipitate (scum)** is formed instead of lather.



Types of Hardness



(a) Temporary Hardness

- Caused by **bicarbonates of Ca & Mg** - $\text{Ca}(\text{HCO}_3)_2$, $\text{Mg}(\text{HCO}_3)_2$
- Also called **Carbonate Hardness**
- Removed simply by **boiling** the water

(b) Permanent Hardness

- Caused by **chlorides & sulphates of Ca & Mg** - CaCl_2 , MgCl_2 , CaSO_4 , MgSO_4
- Also called **Non-Carbonate Hardness**
- Cannot be removed by boiling; needs **lime-soda, zeolite or ion-exchange** process

On boiling, temporary hardness is removed because bicarbonate decomposes and insoluble carbonate settles down:



*Trick to remember: "BiCarb Boils away" - Bicarbonate hardness goes away on **Boiling**. Remember Cl, SO₄ = **Cannot boil off** (permanent).*

Exam Tip: Very frequently asked (PYQ) - "Define hardness and differentiate temporary & permanent hardness" (2-4 marks / short answer, almost every year).

Quick Revision: Hardness = soap does not lather, caused by Ca²⁺/Mg²⁺ salts. Temporary = bicarbonates, removed by boiling. Permanent = Cl-/SO₄(2-) salts, removed by chemical treatment.

Types of Water based on Hardness

Type	Total Hardness (as CaCO_3 ppm)
Soft water	0 - 75 ppm
Moderately hard	75 - 150 ppm
Hard water	150 - 300 ppm
Very hard water	Above 300 ppm

Disadvantages of Hard Water

- Wastes soap (extra soap needed to form lather)
- Forms **scale & sludge in boilers** - reduces heat transfer, may cause boiler explosion
- Unfit for use in dyeing, textile, sugar and paper industries
- Unpleasant taste, may cause laxative effect if very hard

Important: "Why boiler feed water must be softened before use?" - Frequently asked (PYQ) short/long answer question.

Expression of Hardness

Hardness is always expressed / reported in terms of an **equivalent amount of CaCO_3** , because CaCO_3 has a molecular weight of exactly **100** and equivalent weight **50**, which makes calculations very simple.

$$\text{Hardness (as } \text{CaCO}_3) = (\text{Amount of hardness causing salt} \times 50) / (\text{Equivalent wt. of that salt})$$

This is the basis of nearly all hardness numerical problems - convert every salt to its CaCO_3 equivalent and add them up.

Trick: "50 se convert karo" - multiply given salt amount by $(50 \div \text{Eq. wt of salt})$ to get CaCO_3 equivalent.

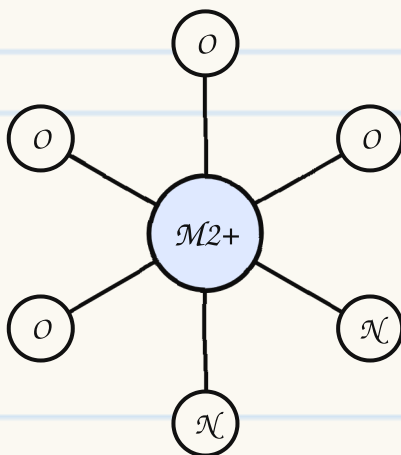
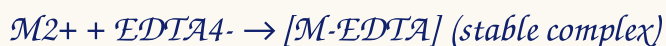
Quick Revision: Hard water wastes soap and damages boilers. All hardness is reported as CaCO_3 equivalent using Eq.wt = 50.

2. EDTA Method (Complexometric Titration)

EDTA = Ethylene Diamine Tetra Acetic Acid. It is a **hexadentate ligand** (has 6 donor atoms - 2 N and 4 O) which can form a very stable ring complex (chelate) with metal ions like Ca^{2+} and Mg^{2+} .

Principle

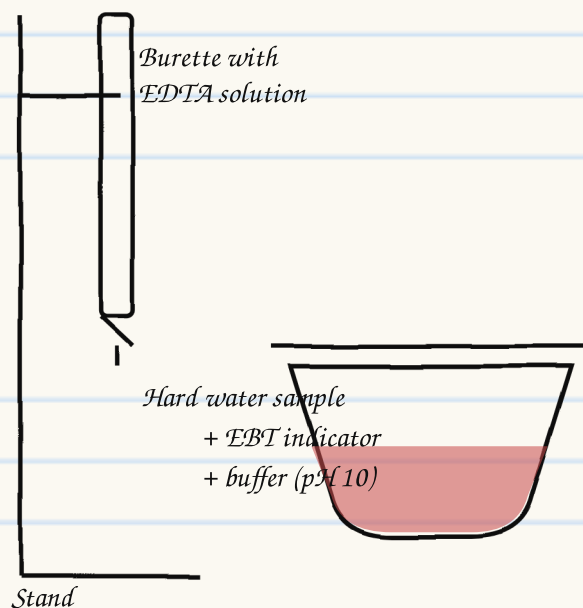
EDTA reacts with Ca^{2+} and Mg^{2+} present in water in **1:1 ratio** to form a stable, colourless complex. The amount of EDTA used (from burette) tells us how much Ca^{2+} / Mg^{2+} (hence hardness) is present in the sample.



EDTA wraps metal ion from 6 sides (hexadentate) - forms stable ring (chelate)

Indicator used: Eriochrome Black-T (EBT)

- Free EBT indicator + buffer (pH 10) → gives **wine red** colour with $\text{Ca}^{2+}/\text{Mg}^{2+}$
- When all metal ions are consumed by EDTA, free EBT is released → colour changes to **pure blue**
- Colour change: **Wine red** → **Blue** = END POINT



Procedure (step-by-step)

1. Pipette out 50 mL (or 100 mL) of hard water sample into a conical flask
2. Add 1 mL buffer solution ($\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$) to maintain $\text{pH} = 10$
3. Add a few drops of EBT indicator $\rightarrow$ solution turns wine red
4. Fill burette with standard EDTA solution
5. Titrate drop by drop with constant shaking, until colour changes from wine red to blue
6. Note burette reading = volume of EDTA used
7. Repeat titration to get concordant readings

$$\text{Hardness (ppm as CaCO}_3) = (V(\text{EDTA}) \times M \times 100 \times 1000) / V(\text{sample})$$

where M = molarity of EDTA (mol/L), $V(\text{EDTA})$ and $V(\text{sample})$ in mL. (Exact constant used depends on standardisation - always shown with unit check in numericals.)

Trick to remember colour change: "Red before, Blue after" - like a sunset (Red) turning into calm Blue sky at the end point.

PYQ Alert: "Describe EDTA method for determination of hardness of water with reaction and indicator used" - asked almost every year (5-7 marks).

Quick Revision: EDTA (hexadentate) + $\text{Ca}^{2+}/\text{Mg}^{2+} \rightarrow$ stable complex (1:1). EBT indicator: wine red $\rightarrow$ blue at end point, buffer $\text{pH} = 10$ used.

Two Titrations usually performed

Titration	What it gives
Sample water directly with EDTA (using EBT, pH 10)	Total Hardness ($\text{Ca}^{2+} + \text{Mg}^{2+}$)
Sample water + NaOH (removes Mg as $\text{Mg}(\text{OH})_2$) titrated with EDTA (using Murexide indicator)	Calcium hardness only

Magnesium hardness = Total hardness – Calcium hardness

Advantages of EDTA method (over old soap method)

- Very accurate and quick
- Sharp colour change gives precise end point
- Gives Ca and Mg hardness separately
- Suitable for boiler feed water testing in industries

Common mistake to avoid: Students forget to add **buffer solution** before adding EBT indicator - without buffer (pH 10), the colour change is not sharp. Always mention this step in exam!

Standardisation of EDTA

EDTA solution is standardised against a standard hard water sample prepared by dissolving a known weight of pure CaCO_3 in dilute HCl . This gives the exact '**1 mL EDTA = X mg CaCO_3** ' factor used later in all calculations.

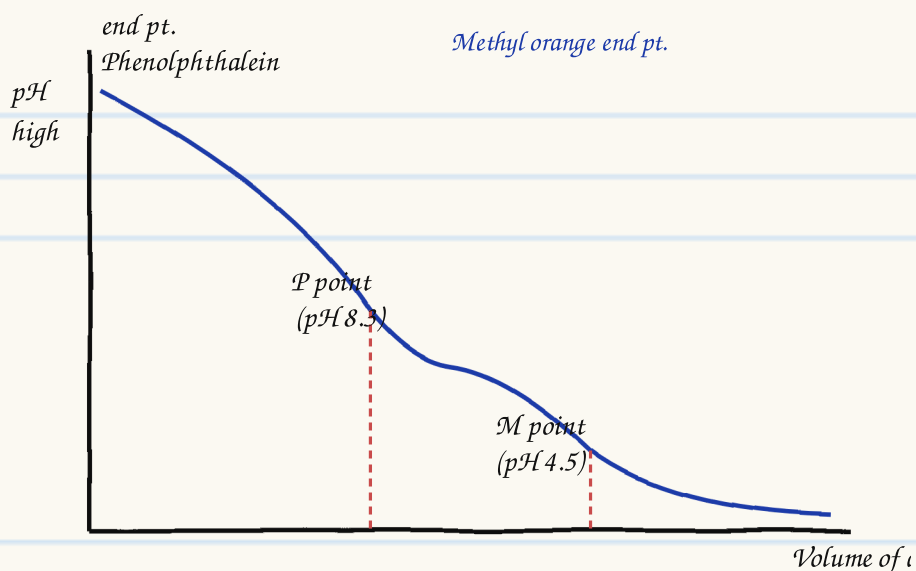
Quick Revision: Two titrations (EBT for total hardness, Murexide after NaOH for Ca-hardness) → Mg hardness by difference. EDTA is first standardised against pure CaCO_3 solution.

3. Alkalinity of Water

Definition: Alkalinity is the capacity of water to **neutralise acids**. It is due to presence of **hydroxide (OH^-)**, **carbonate (CO_3^{2-})** and **bicarbonate (HCO_3^-)** ions in water (any one, two, or all can be present, but never OH^- and HCO_3^- together).

Determination - Two indicators used

- **Phenolphthalein indicator** - colourless below pH 8.3, pink above pH 8.3. Titration with acid up to colour change (pink to colourless) gives value '**P**'.
- **Methyl orange indicator** - titration continued further till pH 4.5 gives total value '**M**' (Total alkalinity, $P + \text{rest}$).



Relationship between P and M (Rule Table)

Let P = phenolphthalein alkalinity (titre value), M = total (methyl orange) alkalinity

Result of titration	OH^-	CO_3^{2-}	HCO_3^-
$P = 0$	0	0	M
$P = M$	P	0	0
$P < 1/2 M$	0	$2P$	$M - 2P$
$P = 1/2 M$	0	$2P$	0
$P > 1/2 M$	$2P - M$	$2(M - P)$	0

Trick: "Compare P with Half of M " - always compare P to $M/2$ first, that decides which row applies. ($P=0 \rightarrow$ only HCO_3^- , $P=M \rightarrow$ only OH^- .)

PYQ Alert: The P/M alkalinity rule table + a small numerical based on it is asked very frequently (5-8 marks question).

Quick Revision: Alkalinity = acid-neutralising capacity, from OH^- , CO_3^{2-} , HCO_3^- . Found using phenolphthalein (P) and methyl orange (M) titration; use the P vs $M/2$ rule table to find which ions are present.

4. Units of Hardness

Hardness is always finally expressed as an equivalent amount of CaCO_3 , but it can be reported in different units:

Unit	Meaning
ppm (parts per million)	mg of CaCO_3 equivalent hardness per 10^6 (million) parts (by weight) of water
mg/L (milligram per litre)	mg of CaCO_3 equivalent hardness present per litre of water
Degree Clark ($^\circ\text{Cl}$)	Number of grains (1 grain = $1/70$ gram) of CaCO_3 equivalent hardness per gallon (imp.) of water = parts of CaCO_3 per 70,000 parts of water
Degree French ($^\circ\text{Fr}$)	Parts of CaCO_3 equivalent hardness per 10^5 (1,00,000) parts of water
Epm (equivalents per million)	Number of equivalents of CaCO_3 hardness per 10^6 parts of water

Important relation (since density of water $\sim 1 \text{ g/mL}$):

$$1 \text{ ppm} = 1 \text{ mg/L}$$

Conversion Table (very important for numericals)

1 ppm	= 1 mg/L
1 ppm	= 0.1 $^\circ\text{Fr}$ (French)
1 ppm	= 0.07 $^\circ\text{Cl}$ (Clark)
1 $^\circ\text{Cl}$	= 14.3 ppm
1 $^\circ\text{Fr}$	= 10 ppm
1 epm	= 50,000 ppm

Trick: "Clark = costliest (biggest number, 70,000), French = fair (1,00,000... wait, actually French base is 1 lakh, Clark base is 70,000)." Simple way: remember only **ppm = mg/L** and derive others using 70,000 (Clark) and 1,00,000 (French) as base parts of water.

PYQ Alert: "Define various units of hardness with inter-conversion" - short answer, asked often; also used inside bigger numericals.

Quick Revision: Units: ppm, mg/L (same), Degree Clark (per 70,000 parts), Degree French (per 1,00,000 parts), epm. Always convert to ppm first before solving numericals.

5. Numerical Problems

Type A: Hardness from salts present (mg/L basis)

Q1. A water sample contains: $\text{Mg}(\text{HCO}_3)_2 = 73 \text{ mg/L}$, $\text{CaCl}_2 = 111 \text{ mg/L}$, $\text{MgSO}_4 = 60 \text{ mg/L}$. Calculate total hardness as CaCO_3 .

Step 1: Use formula: $\text{CaCO}_3 \text{ equivalent} = (\text{mass of salt} \times 50) / (\text{Eq. wt. of salt})$

Salt	Mol. Wt.	Eq. Wt.	Mass (mg/L)	$\text{CaCO}_3 \text{ equiv. (mg/L)}$
$\text{Mg}(\text{HCO}_3)_2$	146	73	73	$73 \times 50 / 73 = 50.0$
CaCl_2	111	55.5	111	$111 \times 50 / 55.5 = 100.0$
MgSO_4	120	60	60	$60 \times 50 / 60 = 50.0$

Step 2: Total Hardness = $50.0 + 100.0 + 50.0 = 200 \text{ mg/L (ppm) as CaCO}_3$

Trick: Always make a small table: Salt | Mol.Wt | Eq.Wt(=Mol.Wt/basicity or valency) | given mass | (mass $\times$ 50/Eq.wt). Then just add the last column!

Type B: EDTA Titration Numerical

Q2. 0.28 g of pure CaCO_3 was dissolved in HCl and made up to 1 litre with distilled water (standard hard water). 50 mL of this standard hard water required 45 mL of EDTA solution. Also 50 mL of the given hard water sample required 10 mL of same EDTA. Find hardness of the given water sample as CaCO_3 equivalent (ppm).

Step 1: Strength of standard hard water = $0.28 \text{ g/L} = 280 \text{ mg/L as CaCO}_3$

Step 2: In 50 mL standard hard water, mass of $\text{CaCO}_3 = 280 \times 50 / 1000 = 14 \text{ mg}$, needing 45 mL EDTA

$$1 \text{ mL EDTA} = 14 / 45 = 0.311 \text{ mg CaCO}_3$$

Step 3: For given sample: 10 mL EDTA used for 50 mL sample

$\text{CaCO}_3 \text{ equivalent in 50 mL sample} = 10 \times 0.311 = 3.11 \text{ mg}$

Hardness = $(3.11 \text{ mg} / 50 \text{ mL}) \times 1000 = 62.2 \text{ mg/L (ppm) as CaCO}_3$

Exam Tip: In EDTA numericals, ALWAYS first find "1 mL EDTA = ? mg CaCO_3 " using the standard hard water data, then apply the same factor to sample data.

Type C: Alkalinity Numerical (P and M titration)

Q3. 100 mL of water sample required 8 mL of $N/50$ HCl using phenolphthalein indicator (P) and a further 12 mL of same acid using methyl orange indicator (to reach M, so total = 20 mL). Report the type and amount of alkalinity present.

Step 1: $P = 8 \text{ mL}$, Total (M) = $8 + 12 = 20 \text{ mL} \Rightarrow M/2 = 10 \text{ mL}$

Step 2: Compare: $P (8) < M/2 (10) \Rightarrow$ from the rule table: $OH^- = 0$, $CO_3^{2-} = 2P$, $HCO_3^- = M - 2P$
 CO_3^{2-} (in mL acid) = $2 \times 8 = 16 \text{ mL}$; HCO_3^- (in mL acid) = $20 - 16 = 4 \text{ mL}$

Step 3: Convert volumes of $N/50$ HCl to $CaCO_3$ equivalent (mg/L):

$$\text{Alkalinity (ppm as } CaCO_3) = (\text{mL of acid} \times \text{Normality of acid} \times 50 \times 1000) / (\text{mL of sample})$$

For CO_3^{2-} part: $(16 \times (1/50) \times 50 \times 1000) / 100 = \mathbf{160 \text{ ppm as } CaCO_3}$

For HCO_3^- part: $(4 \times (1/50) \times 50 \times 1000) / 100 = \mathbf{40 \text{ ppm as } CaCO_3}$

Total Alkalinity = $160 + 40 = \mathbf{200 \text{ ppm as } CaCO_3}$ (present as carbonate + bicarbonate, no hydroxide)

Trick: Solve alkalinity numericals in 3 clean steps every time: (1) find P and M, (2) apply P vs M/2 rule table for ions, (3) convert each ion's acid-volume to ppm using the standard formula.

Type D: Unit conversion numerical

Q4. Express a hardness of 250 ppm in terms of degree Clark and degree French.

Degree French = $\text{ppm} \times 0.1 = 250 \times 0.1 = \mathbf{25^\circ Fr}$

Degree Clark = $\text{ppm} \times 0.07 = 250 \times 0.07 = \mathbf{17.5^\circ Cl}$

PYQ Alert: Numericals of Type A, B and C together make up almost 60-70% of numerical questions asked in RGPV Unit-1 exams - practice all three types thoroughly!

Quick Revision: Salts $\rightarrow CaCO_3$ eq. table method (Type A). EDTA $\rightarrow$ find "1 mL EDTA = ? mg $CaCO_3$ " factor first (Type B). Alkalinity $\rightarrow$ find P, M, apply rule table, convert to ppm (Type C). Simple unit conversions using 0.1 (French) and 0.07 (Clark) factors (Type D).

Unit 1 - Final Quick Revision Sheet

Hardness

- Soap doesn't lather due to $\text{Ca}^{2+}/\text{Mg}^{2+}$
- Temporary = bicarbonates, removed by boiling
- Permanent = Cl^-/SO_4 salts, chemical removal
- Always expressed as CaCO_3 equivalent (Eq.wt=50)

EDTA Method

- EDTA = hexadentate ligand, 1:1 with metal ion
- Indicator: EBT, buffer pH 10
- Colour change: wine red $\rightarrow$ blue
- Ca hardness via Murexide (after NaOH addition)

Alkalinity

- Due to OH^- , CO_3^{2-} , HCO_3^- (never OH^- + HCO_3^- together)
- P = phenolphthalein value; M = total (methyl orange) value
- Compare P with $M/2$ to find which ions present

Units

- 1 ppm = 1 mg/L
- 1 ppm = 0.1 $^\circ\text{Fr}$ = 0.07 $^\circ\text{Cl}$
- 1 epm = 50,000 ppm

Most Important (High-weightage) Questions:

1. Define hardness. Differentiate temporary and permanent hardness. (2018,19,22,23)
2. Describe EDTA method for determination of hardness with reactions and indicator. (Almost every year)
3. What is alkalinity? Explain determination of P and M alkalinity with rule table. (2019,21,23)
4. Numerical: find hardness from given salts / EDTA titre / alkalinity titre (Every year, 1 numerical guaranteed)
5. Explain units of hardness with inter-conversion. (Frequently asked short question)

One-line memory chain: "Hard water $\rightarrow$ Ca/Mg salts $\rightarrow$ No lather $\rightarrow$ Boil removes temporary $\rightarrow$ EDTA (wine red $\rightarrow$ blue) measures total $\rightarrow$ Alkalinity (P, M) tells $\text{OH}^-/\text{CO}_3^{2-}/\text{HCO}_3^- \rightarrow$ Everything reported as CaCO_3 (Eq.wt 50)."